

Alpha-One Intermittency: Complete Periodic Coding and Exact First Returns

HCS-C381 theorem and reproducibility package

5 September 2026

Abstract

We fix the alpha-one intermittent map with its partition endpoint assigned to the left branch. Every binary periodic word then gives exactly one based periodic point, so the fixed-point count is 2^n at every iterate. The all-zero orbit is neutral; every other periodic orbit is expanding. The first return to $(1/2, 1]$ has an explicit countable family of inverse branches, including all literal interval endpoints. An exact reciprocal recurrence gives the unnormalized Lebesgue return tail $1/(4n)$ and the corresponding infinite mean under normalized Lebesgue measure on the base. Nonneutral primitive cycles correspond bijectively to induced cycles, preserving the total multiplier and original time. This first version closes coding, boundary conventions, and exact inducing; it makes no stationary-density claim at the alpha-one endpoint.

中文摘要

本文固定一阶间歇映射，并把分割端点明确归入左支。

每个二进制周期字都对应唯一的有基点周期轨道，因此任意迭代的固定点数精确等于二的相应幂。

全零轨道为中性，其余周期轨道至少经过一次右支并具有严格扩张的总乘子。

对右半区间的首次返回具有可显式写出的可数逆分支，半开半闭端点约定保证这些分支确实构成返回划分。

由左逆迭代的倒数递推得到未归一化勒贝格返回尾的尖锐系数四分之一，基区间归一化后的返回时间均值因此发散。

每条非中性本原轨道与诱导本原轨道一一对应，总乘子不变，原始时间等于各次返回时间之和。

本轮封闭完整编码、端点和首次返回，不把参考勒贝格分布的结论扩张为尚未证明的平稳密度定理。

Keywords: intermittent map; periodic coding; neutral fixed point; first return; endpoint convention; return tail
关键词：间歇映射；周期编码；中性固定点；首次返回；端点约定；返回尾

1 A fixed endpoint and the complete periodic census

Fix

$$T(x) = \begin{cases} x + 2x^2, & 0 \leq x \leq 1/2, \\ 2x - 1, & 1/2 < x \leq 1. \end{cases} \quad g(z) = \frac{\sqrt{1+8z}-1}{4}, \quad h(z) = \frac{1+z}{2}. \quad (1)$$

The endpoint $1/2$ belongs to the left branch, so $T(1/2) = 1$. Its assignment changes a countable boundary orbit set relative to the other common convention and will matter for literal return intervals. The complex square root is chosen with positive real part on $\operatorname{Re} z > 0$. The neutral point is zero. Original time counts one application of T .

Theorem 1 (All periodic words, with the neutral orbit separated). *For every integer $n \geq 1$, T^n has exactly 2^n fixed points. Binary length- n words give a bijection with these based points. The number of primitive cycles of period n is*

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) 2^{n/d}. \quad (2)$$

Here μ is the Möbius function. Only the all-zero orbit is neutral. Every other periodic orbit has positive multiplier at least two. Least word period and least orbit period coincide.

Proof. On $[0, 1]$, the maps g, h are increasing, $0 < g' \leq 1$, and $h' = 1/2$. Any inverse composition containing h is a contraction with Lipschitz constant at most $1/2$, so it has exactly one fixed point. The all- g composition has only zero because $g(x) < x$ for positive x ; the all- h composition has fixed point one.

Before reading forward itineraries, exclude boundary ambiguity. If an intermediate point of a mixed inverse cycle were zero, the inverse step producing it would be g with zero input. Iterating this implication around the cycle forces every branch to be g . If it were one, the step would be h with input one, forcing the all- h cycle. Thus mixed cycles avoid zero and one. They also avoid $1/2$, which can arise only as $g(1)$ or $h(0)$. Their forward

itineraries are therefore unambiguous and agree with the prescribed branch composition. Two different based words cannot produce the same point because the deterministic forward orbit has only one itinerary. Conversely every fixed point solves the equation for its inverse word. This proves the complete count and identifies least periods.

Möbius inversion of the primitive-power decomposition gives (2). Every nonzero periodic word visits the right branch, where $T' = 2$, while each left derivative is $1 + 4x \geq 1$. Multiplying these positive derivatives gives the multiplier statement. Under a repetitions, the period is multiplied by a and the multiplier becomes Λ^a . $\square$

Each branch preserves orientation. No time-reversal symmetry is declared for this noninvertible interval map.

2 Every first-return branch and the sharp tail

Set $Y = (1/2, 1]$, let τ be the first positive return time to Y , and write $R = T^\tau$. Define

$$h_n(z) = \frac{1 + g^{n-1}(z)}{2} \quad (n \geq 1), \quad a_m = g^m(1/2) \quad (m \geq 0). \quad (3)$$

Theorem 2 (Complete return partition and reference-measure tail). *Every point of Y has a finite first return, and*

$$\{y \in Y : \tau(y) = n\} = h_n(Y), \quad R \circ h_n = \text{id}_Y. \quad (4)$$

These left-open, right-closed intervals partition Y . For $n \geq 1$,

$$\text{Leb}\{y \in Y : \tau(y) > n\} = \frac{a_{n-1}}{2} \sim \frac{1}{4n}. \quad (5)$$

Under normalized Lebesgue measure on Y , the tail is asymptotic to $1/(2n)$ and its mean is infinite.

Proof. After a point of Y takes the right branch it is positive. While it remains in $(0, 1/2]$, the left branch increases it strictly. If it never escaped, it would converge to a positive fixed point of $x + 2x^2$, which is impossible. The inverse of a return of length n first reverses $n - 1$ left steps and then one right step, giving (3). The preceding inverse left iterates lie in $(0, 1/2]$, which proves the first-return assertion. Moreover $g(1) = 1/2$ gives $h_{n+1}(1) = h_n(1/2)$. The left-branch assignment at $1/2$ makes the shared upper endpoints return on the claimed iterate. These adjacent intervals exhaust Y as their left endpoints tend to $1/2$.

Their endpoints give the exact tail in (5); for $n = 1$ it is $(1/2, 3/4]$, of length $1/4$. The recurrence $a_{m-1} = a_m + 2a_m^2$ gives

$$\frac{1}{a_m} - \frac{1}{a_{m-1}} = \frac{2}{1 + 2a_m}, \quad \frac{1}{2m+2} \leq a_m \leq \frac{1}{m+2}. \quad (6)$$

The increment lies between one and two. Using the upper bound in $2/(1+2a_m) = 2 - 4a_m/(1+2a_m)$ and summing yields $a_m^{-1} = 2m + O(\log(m+2))$. Therefore $a_m = (2m)^{-1} + O(\log m/m^2)$ and (5) follows. Normalization doubles the tail, and the harmonic-series comparison gives the infinite mean. This is a reference-Lebesgue statement; no stationary induced density has been assumed. $\square$

Corollary 1 (Primitive induction preserves the original clock). *Nonneutral primitive T cycles correspond bijectively to primitive R cycles. If an induced cycle has return period r_p , return times $n_1, \dots, n_{r_p}$, and total multiplier Λ_p , its original period is $N_p = \sum_j n_j$ and its multiplier remains Λ_p .*

Proof. Every nonneutral periodic word contains a right symbol and hence visits Y . Successive gaps between such visits determine the induced itinerary. Conversely (3) reconstructs the original cycle. Changing the chosen visit rotates the induced word. Thus no additional multiplicity equal to the number of returns is introduced. A smaller original or induced period would force a smaller period on the other side. $\square$

The unweighted zeta and its neutral-point removal are consequently

$$Z_T(z) = \frac{1}{1-2z}, \quad Z_{T,\text{nonneutral}}(z) = \frac{1-z}{1-2z}, \quad |z| < 1/2. \quad (7)$$

Indeed the first is the exponential series of the exact fixed counts; the second removes the single primitive fixed point. The elementary renewal identity $1 - \sum_{n \geq 1} z^n = (1-2z)/(1-z)$ is consistent with the same original clock. These statements concern unweighted counting.

Literature ownership and claim boundary

The intermittent-map lineage is represented by Liverani, Saussol and Vienti [1]. The alpha-one endpoint is treated by the explicit calculations here; no finite stationary-measure theorem is imported from a different parameter range. Rugh's work [2] establishes the classical connection between intermittency and regularized Fredholm determinants on carefully specified function spaces. The present package neither claims to discover inducing nor replaces those function-space qualifications. Its increment is the connected audit of one literal endpoint convention, complete primitive coding, sharp reference return tail, and the exact original-time return correspondence. No worldwide novelty or publication-acceptance claim is made.

References

- [1] C. Liverani, B. Saussol and S. Vienti, "A probabilistic approach to intermittency," *Ergodic Theory and Dynamical Systems* 19 (1999), 671–685, doi:10.1017/S0143385799133856.
- [2] H. H. Rugh, "Intermittency and Regularized Fredholm Determinants," preprint (1996), arXiv:chao-dyn/9610011.

Round zero: complete periodic coding and exact first returns